U.G.C. NET Exan
COMPUTEI
(Solved with Explar

1. Which of the following is NOT logically equivalent to P → Q?

(a) ~P v Q (b) ~ Q → ~P

(c) ~ (P ^ ~ Q) (d) ~ Q → P

Ans. (d) : By using truth table

P	Q	~P	~Q	P→Q	~P∨Q	~Q→~P	~Q→P	~(P^~Q
T	T	F	F	T	T	T	T	T
T	F	F	T	F	F	F	T	F
F	T	T	F	T	T	T	T	T
F	F	T	T	T	T	T	F	T

So by truth table option (d) is correct.

2. Seven people enter a lift. The lift stops only at three floors (which are unspecified). At each of the three floors no one enters the lift, but at least one person leaves the lift. After the three floor stops the lift is empty in how many ways this can happen?

(a) 1844 (b) 2187 (c) 1806 (d) 343

Ans. (c) : According to the question there are many possible combinations in that case.

i.e. 5 – 1 – 1, 4 – 2 – 1, 4 – 1 – 2, 3 – 3 – 1,
3 – 2 – 2, 3 – 1 – 3, 2 – 4 – 1, 2 – 3 – 2,
2 – 2 – 3, 2 – 1 – 4, 1 – 5 – 1, 1 – 4 – 2,
1 – 3 – 3, 1 – 2 – 4, 1 – 1 – 5.

There are just 4 possible patterns,
5 – 1 – 1, 4 – 2 –1, 3 – 3 – 1 and 3 – 2 – 2

In each possible way you need to find the permutations.
Computer (Choose people) × (Permute patterns),

For the 3 – 3 – 1 pattern, $\binom{7}{3}\binom{4}{3}(!)\times\frac{3!}{2!}$

By computing each pattern and adding up then.
The answer is 1806.

3. The coefficients of $x^2y^3z^2$ in the expansion of $(x+y+z)^7$ is

(a) C(7, 3) * C(5, 2) (b) C(7, 2) * C(5, 3)

(c) C(7, 3) * C(7, 2) (d) C(5, 3) * C(5, 2)

Ans. (b) :
General term of $(x+y+z)^7$

$$=\frac{7!}{p!q!r!}x^p\times y^q\times z^r$$

So coefficient of $x^2y^3z^2$

$$=\frac{7!}{2!3!2!}\times 1^2.1^3.1^2=\frac{7!}{2!3!2!}=210$$

So option (b) is correct

4. What is the minimum number of students in discrete mathematics class to be sure that at least six will receive the same grade, if there are seven possible grades : S, A, B, C, D, P and F?

(a) 7 (b) 13 (c) 36 (d) 14

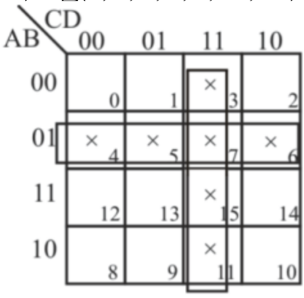
Ans. (c) :
The given data
Seven possible grades = {S, A, B, C, D, P, F}
n = 7
get the same grade
K + 1 = 6
K = 5
By pigeonhole principle
The minimum number of student = kn+1
= 5 × 7 + 1 = 35 + 1 = 36
So option (c) is correct.

5. In 16-bit 2's Complementation representation, the decimal number 28 is
(a) 1111 1111 0001 1100
(b) 1111 1111 1111 0100
(c) 1111 1111 1110 0100
(d) 1111 1111 1110 0011

Ans. (c) : Given decimal number
$(28)_{10} = (11100) = (0000\ 0000\ 0001\ 1100)_2$
2's complement of 0000 0000 0001 1100
= 1111111111100100
Note : Start reading the bits from LSB (right hand side) and write it unless first 1 is encounter leave the first 1 as it is and complement the remaining bits.

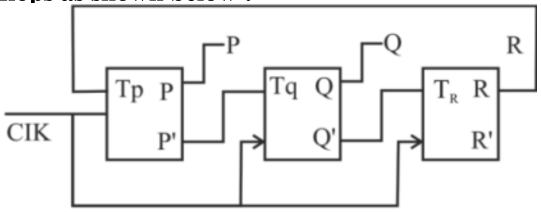
6. Simplify the following using K-Map F(A, B, C, D) = ∑(3, 4, 5, 6, 7, 11, 15)
(a) AB + C'D
(b) A'B + CD
(c) AC + BD
(d) A'B + B'C

Ans. (b) : Given K-map
F(A, B, C, D) = ∑(3, 4, 5, 6, 7, 11, 15)



So simplified Expression = A'B + CD

7. Consider a 3-bit counter, designed using T flip flops as shown below :



Assume the initial state of the counter given by PQR as 000, what are the next three states?
(a) 011, 101, 000
(b) 001, 010, 111
(c) 011, 101, 111
(d) 001, 010, 000

Ans. (a) : From the above counter-

$T_P = R$, $T_Q = \bar{P}$, $T_R = \bar{Q}$ and PQR = 000

Present Input			**Next state**		
$T_P = R$	$T_Q = \bar{P}$	$T_R = \bar{Q}$	P^+	Q^+	R^+
0	1	1	0	1	1
1	1	0	1	0	1
1	0	1	0	0	0

Where $P^+ = P \oplus P$, $Q^+ = \bar{P} \oplus Q$, $R^+ = \bar{Q} \oplus R$

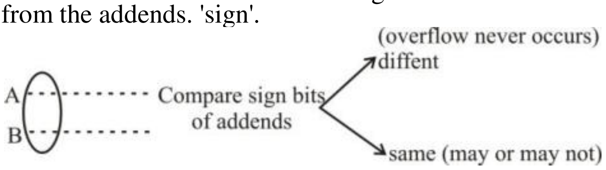
So, next three states are- 011, 101, 000

8. Let R1 and R2 be two 4-bit registers that store numbers in 2's complement form. For the operation R1+ R2 which one of the following values of R1 and R2 gives an arithmetic overflow?

(a) R1 = 1011 and R2 = 1110
(b) R1 = 1100 and R2 = 1010
(c) R1 = 0011 and R2 = 0100
(d) R1 = 1001 and R2 = 1111

Ans. (b) : If two numbers are represented using 2's - complement with n-bits

Condition for overflow : Overflow occurs if the signs of addends are the same and the sign of sum is different from the addends. 'sign'.



Note : Same and then different ⇒ overflow

Here, In this particular question :

Here in all option sign bits of addends are same ⇒ overflow many occur

	(A)	(B)	(C)	(D)
R1	1011	1100	0011	1001
R2	1110	1010	0100	1111

(Same in all options)

	(A)	(B)	(C)	(D)
R1	1011	1100	0011	1001
R2	1110	1010	0100	1111
	1001	0110	0111	1000

different

only (b) option, sum's sign if different from addends' sign ⇒ overflow.

It should be roted that while perfarming the addition carry bit generated will be discasded.

9. In the context of Binding which one of the following is incorrect?

(a) Static binding is done at Compile Time
(b) Absolute addresses are known at Load Time
(c) Internal functions are known at link time
(d) Target of a function is known at runtime

Ans. (c) : Static binding is done at compile time. This is correct. Static binding also known as early binding, occurs the competation phase. The compiler resolves the method of function calls at compile time and the binding is fixed.

- Absolute addresses are known as load time This is also correct. In a system with static memory allocation, the addresses of variables and functions are determined and fixed during the load time of the program.

- Internal functions are known as compile time. This is the incorrect statement. Internal functions also known as static functions have their scope limited to the file they are defined in they are not known outside of the file they are delcared in at compile time but they are still knwon within file during the linking phase.

So option (c) is correct.

10. Given int arr[3][4][2] = {{{2, 4}, {7, 8}, (3, 4}, {5, 6}}, {{7, 6}, {3, 4}, {5, 3}, {2, 3}}}, {{8, 9}, {7, 2}, {3, 4}, {5,1}}}};
Which one of the following can be used to refer to element 1 in the above 3-d array?

(a) arr[3][4][2]
(b) *(*(*(arr+2)+3)+1)
(c) *(arr+2) + *(arr+3) + *(arr+1)
(d) arr[2][3][2]

Ans. (b) : The given code

int arr[3][4][2] = {{2, 4}, {7, 8}, {3, 4}, {5, 6}}
{{7, 6}, {3, 4}, {5, 3}, {2, 3}}
{{8, 9}, {7, 2}, {3, 4}, {5, 1}}}}

To refer to element 1 in the given 3- dimensional array, you should uses option (b)

((*(arr + 2) + 3) +1)

Let's break this expression step-by-step

1. *(arr+2) : This accesses the 3rd element (index 2) in the outermost dimensional of the array. This given a 2-dimensional of the array. This given a 2-dimensional array {{8,9}, {7,2}, {3,2}, {3,4}, {5,1}}
2. *(*(arr+2)+3) : Now we move to the 4th element (index 3) in the second dimensional of the 2-dimensional array obtained in the previous step.

This given us {5, 1}

3. *(*(*(arr+2) + 3) +1) : Finally we access the 2nd element (index 1) in the last 1-dimensional array obtained in the previous step, which given us the desired element.

So option (b) is correct.

11. Allowing different derived classes to be used interchangeably though the interface provided by a common base class is called _____

(a) Derived polymorphism
(b) Compile time polymorphism
(c) Static polymorphism
(d) Run time polymorphism

Ans. (d) : Run time polymorphism- This is achieved through the use of virtual functions and is a fundamental concept in object oriented programming. At runtime the appropriate derived class's implmentation is called based on the actual type of the object being referred to through a pointer or reference to the base class. This allows for dynamic dispatch and enables the program to behave polymorphically choosing the appropriate method implementation based on the actual runtime type of the object. So option (d) is correct.

12. In ______ cruve the last and the first vertices of the polygon lie on the curve and all other vertices define the derivatives order and shape of the curve.

(a) B-Spline Curve (b) Boxline curve
(c) Bezier curve (d) Gaussain curve

Ans. (c) : In a Bezier curve, the first and the last vertices of the polygon lie on the curve, and all the other vertices (also called control points) define the derivatives, order and shape of the curve. Bezier curves are commonly-aided design (CAD) applications for creating smooth curves and surfaces. They are define by Bernstein polynomials and offer precise control over the shape of the curve by adjusting the positions of the control points.

13. A ______ operation combines the rows of the two relations and outputs a new relation that has both input relations rows in it.

(a) Difference (b) Union

(c) Certesian product (d) Project

Ans. (b) : The operation that combines the rows of two relations and outputs a new relation that has both input relation's rows in its is the "Union operation.

So the option (b) is correct.

14. Consider the relation R(P, Q, R, S, T, U, V, W, X, Y) and FDs :

{P, Q, S} → T

R → Y

{P, Q} → V

{R, X} → X

Q → U

V → {W, X}

Which of the following is a candidate key?

(a) {P, Q, R, S} (b) {R, Q, X}

(c) {V, X} (d) None of the above

Ans. (a) : Given the function dependencies (FDs)

1. (P, Q, S) → T
2. R → V
3. {P, Q} → V
4. (R, X) → X
5. Q → U
6. V → {W, X}

Let's check if all attributes are covered by {P, Q, R, S}

- T : T is covered by (P, Q, S) → T
- U : U is covered by Q → V
- V : V is covered by {P, Q} → V
- W : W is not directly determined by any candidate key.
- X : X is covered by {R, X} → X and V → {W, X}
- Y : Y is covered by (R) → Y

Since all attributes are covered {P, Q, R, S} is a candidate key.

So option (a) is correct.

15. Which RAID level is appropriate for storing the log file of a ticket booking system where a huge database update is required?

(a) Level 0 (b) Level 1

(c) Level 2 (d) Level 3

Ans. (b) : RAID (Redundant Array of Independent Disks) Level 1 provided mirroring, meaning that data is duplicated across two or more drives. In this configuration, all data written to one drive is also written to another drive in real-time. This redundancy ensures data integrity and availability making it a suitable choice for storing critical log file that must be preserved and protected during a huge database update.

So option (b) is correct.

16. If we convert a decision tree to a set of logical rules then :

(a) The internal nodes in a branch are connected by AND and branches by AND

(b) The internal nodes in a branch are connected by OR and branches by OR

(c) The internal nodes in a branch are connected by AND and branches by OR

(d) The internal nodes in a branch are connected by OR and branches by AND

Ans. (c) : The correct statement is : The internal nodes in a branch are connected by "AND" and branches by "OR".
When converting a decision tree into a set of logical rules, each path from the root to a leaf in the decision tree represents a conjunction of conditions, and the different paths branches are combined using the logical OR operation. Each internal node in a branch represents a condition connected by logical AND. By following this approach you can express the decision tree as a set of logical rules that capture the same decision - making process.

17. Model of protection can be viewed as
(a) Owner matrix (b) Domain matrix
(c) Access Matrix (d) Global matrix

Ans. (c) : Access Matrix - The model of protection that we have been discussing can be viewed as an access matrix in which columns represent different system resources and rows represent different protection domains. Entries within the matrix indicate what access that domain has to that resource.

domain \ object	F1	F2	F3	Printer
D1	read		read	
D2				Print
D3		read	execute	
D4	read write		read write	

This is access matrix graph

18. Process in memory also contain the heap section. What is the use of heap section?
(a) Contains temporary data
(b) Dynamically allocate the memory during process run time
(c) Contains Process priority and pointers to scheduling queues
(d) Contains amount of CPU and real time use.

Ans. (b) : The heap section in memory is crucial for managing dynamically allocate data during a process's runtime. It allow programs to allocate memory as needed, making it possible to handle data with varying sizes and life spans efficiently.
However proper memory management is essential to prevent memory leaks and excessive memory consumption.
So option (b) is correct.

19. How many page faults occur for optimal algorithm for the following string with four frames?
1, 2, 3, 4, 5, 3, 4, 1, 6, 7, 8, 7, 8
(a) 8 (b) 7 (c) 6 (d) 5

Ans. (a) : The given reference string :
1, 2, 3, 4, 5, 3, 4, 1, 6, 7, 8, 7, 8
Number of frames : 4
Let's simulate the algorithm step by step
1. Frame status [–, –, –,–] page fault! Page 1 is loaded into frame 1.
2. Frame status : [1, –, –, –] Page fault!
Page 2 is loaded into frame 2.
3. Frame status : [1, 2, –, –] page fault!
Page 3 is loaded into frame 3.
4. Frame status : [1, 2, 3, –] page fault!
Page 4 is loaded into frame 4.
5. Frame status : [1, 2, 3, 4] page fault!
Page 5 is loaded into frame 1.

6. Frame status : [5, 2, 3, 4] page fault!
   Page 3 is already in a frame so no page fault occurs.
7. Frame status : [5, 2, 3, 4] page fault!
   Page 4 is already in a frame, so no page fault occurs.
8. Frame status : [5, 2, 3, 4] page fault!
   Page 1 is already in a frame, so no page fault occurs.
9. Frame status : [5, 2, 3, 4] page fault!
   Page 6 is loaded into frame 2.
10. Frame status : [5, 6, 3, 4] page fault!
    Page 7 is loaded into frame 3.
11. Frame status : [5, 6, 7, 4] page fault!
    Page 8 is loaded into frame 4.
12. Frame status : [5, 6, 7, 8] page fault!

Now, when Page 7 is requested again, it will be replaced because it's the last occurrence in the reference string that requires a page replacement.

Total page faults : 8

So, for the given reference string and four frames the optimal algorithm result in 8 page faults.

20. Degree of concurrency in thread can be arranged in the manner.

(a) One-to-one > many-to-one > many-to-many
(b) One-to-one > many-to-many > many-to-one
(c) many-to-many > many-to-one>one-to-one
(d) None of the above

Ans. (c) : The correct arrangement of degree of concurrency from highest to lowest is many-to-many > many-to-one > one- to - one.

21. Which of the following is not one of the principles of good coding?

(a) Create unit tests before you begin coding.
(b) Create visual layout that aids understanding.
(c) Refractor the code after you completes the first coding pass.
(d) Write self-documenting code not program documentation.

Ans. (d) : Good principles of coding is

- Create unit tests before you begin coding.
- Create visual layout that aids understanding.
- Refractor the code after you completes the first coding pass.
- DRY (Don't repeat yourself)
- KISS (Keep It short and simple)
- Refactor
- Creation over legacy
- Open/closed

So option (d) is correct.

22. If P is the risk probability, L is the loss, then Risk exposure (RE) is computed as _____

(a) RE = P/L (b) RE = P + L
(c) RE = P * L (d) RE = 2 + (P*L)

Ans. (c) : The correct formula to calculate Risk Exposure (RE) based on the risk probability (P) and loss (L) is

RE = P * L

This formula multiplies the risk probability (P) by the loss (L) to determine the Risk Exposure (RE). It takes into account both the likelihood of an event occurring (Risk probability) and the potential impact or loss associated with that event.

23. Which of the following is a technique used for component wrapping?

(a) Black-box wrapping (b) Clear-box wrapping
(c) Gray-box wrapping (d) White-box wrapping

Ans. (b) : Component Adaptation Techniques-

- White-box wrapping (integration conflicts removed by marking code level modifications to the code)
- Grey box wrapping (used when component library provides a component extension language or API that allows conflicts to be removed or masked)
- Black-box wrapping (requires the introduction of pre- and post-processing at the component interface to remove or mask conflicts)

So clear-box wrapping is a technique used for component wrapping.

24. The cyclomatic complexity metric provides the designer with information regarding the number of _____

(a) Cycles in the program
(b) Errors in the program
(c) Independent logic paths in the program
(d) Statements in the program

Ans. (c) : The cyclomatic metric provides the designer with information regarding the number of linearly independent paths through a program's source code. It is a quantitative measure of the complexity of the complexity of a program is controlflow.

The formula to calculate cyclomatic complexity is as follows.

Cyclomatic complexity (V) = E – N + 2

Where

E = the number of edges in the control flow graph

N = the number of nodes in the control flow graph.

25. In a ______ the left and right sub trees for any given node, differ in height by more than one.

(a) Balanced Binary Tree
(b) Weight balanced binary tree
(c) Height balanced tree
(d) Binary Search Tree

Ans. (*) : an unplaced binary tree is a type of binary tree where the height of the left and right sub trees for any given node given differs by more than one level. UGC has drop this question.

26. FFTs are ______ techniques used to encode digital video and audio information and helps analyse the signal in _____ domain.

(a) Encryption, time
(b) Encryption, frequency
(c) Compression, time
(d) Compression, frequency

Ans. (d) : Fourier Transform is a mathematical concept that can covert a continuous signal from time - domain to frequency - domain. FFTs are compression techniques used to analyze digital video and audio information and help analyze the signal in the frequency domain. So the correct answer is compression, frequency.

27. Given six weights 4, 15, 25, 5, 8, 16 construct a 2-T tree. What is the minimum weighted path length?

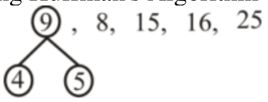
(a) 231 (b) 172 (c) 73 (d) 77

Ans. (b) :

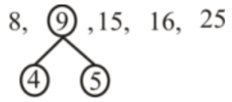
The given weights = 4, 15, 25, 5, 8, 16

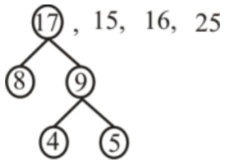
Seat = 4, 5, 8, 15, 16, 25

Using Huffman's Algorithm

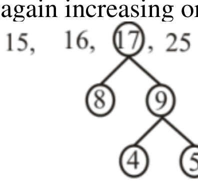
(i) 

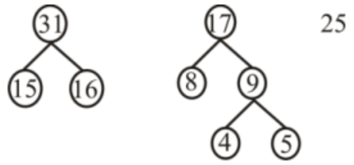
again increasing order.

(ii) 

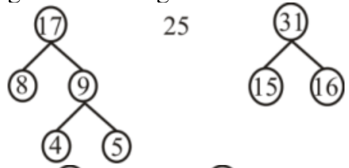
(iii) 

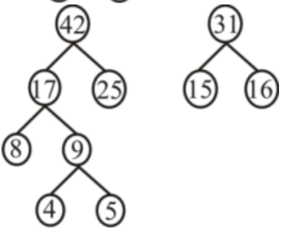
again increasing order

(iv) 

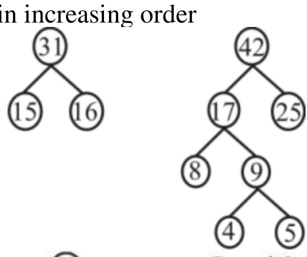


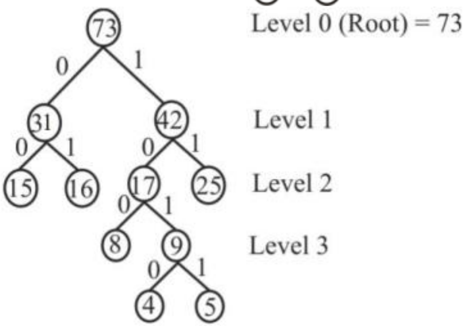
again increasing order

(v) 



again increasing order

(vi) 



Minimum weighted path length

$= 15 \times 2 + 16 \times 2 + 8 \times 3 + 4 \times 4 + 5 \times 4 + 25 \times 2$

$= 30 + 32 + 24 + 16 + 20 + 50$

$= 172$

So option (b) is correct.

28. Suppose we store n keys in a hash table of size $m=n^2$ using a has function h randomly chosen from a universal class of hash functions. Then the probability is _____ that there are any collisions.

(a) greater than 1/5 (b) greater than 1/2
(c) less than 1/2 (d) less than 1/5

Ans. (c) : There are $\binom{n}{2}$ pair of keys that may collide; each pair collides with probability 1/m if n is chosen at random from a universal class of hash function,
Let x be a random variable that counts the number of collisions. When $m=n^2$ the expected number of collisions is

$$E(x) = \binom{n}{2}.\frac{1}{n^2}$$

$$= \frac{n^2 - n}{2}.\frac{1}{n}$$

$$E(x) < \frac{1}{2}$$

So option (c) is correct.

29. In Pumping Lemma for context free languages, to say a language is satisfying pumping lemma, what is the minimum length of 'vx' together, if you consider the string as 'uvwxy'?

(a) 0 (b) 1
(c) 2 (d) n

Ans. (a) : The pumping Lemma for context free languages that for any context - free language L, there exists a constant 'n' (called the pumping length) such that any string S in L, with length at least 'n' can be divded into five pieces u, v, w, x, y satisfying the following conditions.

1. For each $i \geq 0$ the string uv w xy is also in L.
2. $|vwx| \geq 1$ (i.e. the combined length of v and x is at least 1)
3. $|vwx| \leq n$ (i.e. the combined length of v and x is at most the pumping length.)

So in this context, the minimum length of 'vx' to gather (i.e. $|vwx|$) for a language to satisfy the pumping Lemma is 1.
So option (b) is correct.

30. Which of the following is ε-free grammar?

(a) Type-0 (b) Type-1
(c) Type-2 (d) Type-3

Ans. (b) : ε - Free grammar is also known as context sensitive grammar.
$S \rightarrow \in$ it is allow to the starting symbol.
So option (b) is correct by Chomsky hierarachy.

31. While processing a string by an LR parser, the reading of the given string by the parser is ....

(a) From right to left
(b) From left to right
(c) Can be either from left to right or right to left
(d) From the centre of the given string

Ans. (b) : The leading of a given string by an LR parser is from left to right. LR (Left-to-right, right most derivation) parser is a type of shift reduce parser used in parsing computer languages. It starts from the leftmost end of the input string and builds the parse tree in a bottom up manner by repeatedly applying shift and reduce operations until the input is completely parsed.
So option (b) is correct.

32. **What is the minimum and maximum number of initial, dead, unreachable and final states in a valid 'n' state finite automaton, where the language accepted is not 'φ'?**
(a) Minimum : 1, 1, 1, 1; Maximum : 1, (n-1), (n-1), n
(b) Minimum : 1, 0, 0, 0; Maximum : 1, (n–1), (n-1), n
(c) Minimum : 1, 0, 0, n; Maximum: 1, n, (n-1), (n-1)
(d) Minimum : 1, 0, 0, 1; Maximum: 1, (n-1), (n-1), n

Ans. (d) : The minimum and maximum number of initial, dead, unreachable, and final state can be defined as follows.

1. Initial state

- Minimum : There must be at least one initial state for the automation to start its computation So, the minimum number of initial states is 1.
- Maximum : There can be only one initial state in a finite automation having more than one initial state would lead to ambiguity and violate the definition of a finite automation.

2. Dead states (Non-Accepting states)

- Minimum : There can be zero dead state in the automation. A dead state is a state from which no final state can be reached, and it cannot accept any string.
- Maximum : The maximum number of dead states is n-1. This occurs when there is a single accepting state, and all other state are dead states.

3. Unreachable states :

- Minimum : There can be zero unreachable states in the automation, meaning that all states are reachable from the initial states.

  Maximum : The maximum number of unreachable states is n-1. This occurs when there is a single reachable state, and all other states are unreachable from the initial state(s)

4. Final states (Accepting states)

- Minimum : There are must be at least one final state to accept strings in the language. So the minimum number of final states is 1.
- Maximum : The maximum number of final states can be n. This happens when all states in the automation are final states meaning that any state can lead to an accepting state.

So

- Minimum Initial states : 1
- Maximum Initial states : 1
- Minimum Dead states : 0
- Maximum Dead states : n -1
- Minimum Unreachable states : 0
- Maximum Unreachable states : n - 1
- Minimum Final states : 1
- Maximum Final states : n

So option (d) is correct

33. **Which protocol is used to acquire the MAC address of a host whose IP address is known?**
(a) Dynamic Host control protocol
(b) Address Resolution Protocol
(c) Reverse Address Resolution Protocol
(d) User Datagram Protocol

Ans. (b) : Address Resolution Protocol (ARP) is a procedure for mapping a dynamic IP address to permanent physical machine address in a local area network (LAN). The physical machine address is also know as a media access control (MAC) address.

34. In CRC coding if the data word is 111111, divisor is 1010 and the remainder is 110. Which of the following code is true?

(a) 011111101 (b) 001010110
(c) 111111110 (d) 110111111

Ans. (c) : The given data word = 111111
divisor = 1010
remainder = 110
steps to perform CRC on the sender side

- String of n o's is appended to the data unit to be transmitted.
- Where, n→ number of bits in CRC divisor-1
- Binary division is performed of data unit with the CRC divisor.
- The remainder obtained after division is called CRC.
- Append CRC at last of data units.

So append CRC 110 at last of data unit - 111111110
So option (c) is correct.

35. In Go-back-N protocol, the values of S_f(Send window, the first outstanding packet) = 3, S_n (Send window, the next packet to be send) = 7, after receiving the packet with AckNo. 6, the value of S_n is _____

(a) 6 (b) 7
(c) 5 (d) 4

Ans. (b) : Given the values
■ S_f (send window, the first outstanding packet) = 3
■ S_n (Send window, the next packet to be sent) = 7
and after receiving an acknowledgement with ACKNo. 6, it means that packets, 3, 4, 5 and 6 have been successfully received. Thus the sender can safely move its window forward.
The updated value of S_n will be the next packet that needs to be sent. Since packets 3, 4, 5 and 6 have acknowledged the next packet to be sent is packet 7. Therefore, after receiving the acknowledgement with ACK No. 6, the value of S_n becomes 7.
So option (b) is correct.

36. What is the minimum Hamming Distance in a block code if we want to be able to detect up to S errors?

(a) S (b) S+1
(c) 1 (d) 0

Ans. (b) : Hamming distance between two words of the same size is the number of differences between the corresponding bits. It can be calculated by applying the XOR operation.
Formula
To guarantee the detection of up to S errors in all cases the hamming distance in a block code must be
$d \geq S + 1$
hamming distance be minimum
$\therefore$ dmin = S + 1
So option (b) is correct

37. Which search technique is complete and optimal, when h(n) is consitent?

(a) Best First Search (b) Depth First Search
(c) A* Search (d) Breadth First Search

Ans. (c) : When the heuristic function h(n) is consistent (also know as "admissible" or "monotonic", the A* search algorithm becomes both complete and optional.
So option (c) is correct

38. What is the process of capturing inference process as a single inference rule?

(a) Ponens (b) Clauses
(c) Generalized modus ponens (d) Variable

Ans. (c) : All kinds of inference process can be captured as a single inference rule that can be called as generalized modus ponens.

39. Fuzzy logic is a form of ____

(a) Two valued logic (b) Crisp set logic
(c) Binary set logic (d) Many valued logic

Ans. (d) : Fuzzy logic is a form of many-valued logic in which the truth many-valued logic in which the truth value of variable may be any real number between 0 and 1. It is employed to handle the concept of partial truth. Where the truth value may range between completely true and completely false.

40. The 4Ws problem canvas helps in identifying the key elements related to the problem 4Ws problem canvas is a part of ______

(a) Problem scoping (b) Data acquisition
(c) Data modeling (d) Problem Evaluation

Ans. (a) : Problem Scoping : Problem scoping means selecting a problem which we might want to solve using our AI knowledge.
Problem scoping is the process of indentifying the scope the problem (like cause, nature or solution of a problem) that you wish to solve with the help of your project. Who What > Where > Why 4Ws problem canvas is a part of problem scoping.
So option (a) is correct.

41. Consider the following statements :

A. A group can have zero elements.
B. A group cannot have any element which is idempotent except the identity element.
C. An isomorphism from a group <G, *> to <G, *> is called an automorphism
D. If every element in a group is its own inverse then the group must be abelian.

Choose the correct answer about the four statements given above.

(a) B, C and D are correct
(b) Only A and C are correct
(c) Only B and C are correct
(d) All A, B, C and D are correct

Ans. (a) : A group is a non-empty set with an operation that satisfies the associative, has an identity element, and every element of the set has an inverse element.
So statements A is false.
B. A group cannot have any element which is idempotent except the identity element. This is true statement.
C. An isomorphism from a group <G, *> to <G, *> is called an automorphism
This is true statement.
D. If every element in a group is its own inverse then the group must be abelian. This is true statement.
So option (a) is correct.

42. Consider the following statements:

A. The number of labelled binary trees possible with 'n' nodes is (2n)!/(n+1)!
B. The number of binary search trees possible with 'n' nodes is (2n)!/((n+1)!n!)
C. The number of unlabeled binary trees possible with 'n' nodes is C(2n, n)!/(n+1)!
D. The number of simple, undirected and labelled graphs possible with 'n' vertices is $2^{n(n-1)/2}$

Choose the correct answer from the options given below.

(a) A, B and C only correct
(b) A, B and D only correct
(c) B, C and D only correct
(d) A, C and D only correct

Ans. (b) : Binary Trees - A tree whose element have 0 or 1 or 2 children is called a binary tree. Since each element in a binary tree can have only 2 children, we typically name them the left and right child.

- The number of binary search trees possible with 'n' nodes is (2n)!/((n+1)!n!)
- The number labeled binary search trees possible with 'n' nodes is (2n)!(n+1)!
- The number of simple, undirected and labeled graphs possible with n nodes can be calculated using the catalan number formula.

Cn = (2n)!/[n+1!*n!]

So statement C is false.

So option (b) is correct.

43. The relation R = {(1, 3), (1, 1), (3, 1), (1, 2), (3, 3), (4, 4)} defined in (1, 2, 3, 4) is _____

(a) Reflexive, Symmetric but not Transitive
(b) Symmetric anti-symmetric and transitive
(c) Asymmetric, anti-symmetric and not transitive
(d) Not reflexive, not symmetric and not transitive

Ans. (d) : The given relation

R = {(1, 3), (1, 1), (3, 1), (3, 1), (1, 2), (3, 3), (4,4)}

Check Reflexive

If the relation is reflexive

then (a, a) $\in$ R for every a$\in$ {1, 2, 3, 4}

Since (1, 1) (3, 3) (4, 4) $\in$ R but (2, 2) $\notin$ R

$\therefore$ R is not reflexive.

Check symmetric

To check whether symmetric or not

If (a, b) $\in$R then (b, a) $\in$ R

Here (1, 2) $\in$R, but (2, 1) $\notin$R

$\therefore$ R is not symmetric

Check transitive

To check whether transitive or not

If (a, b) $\in$R &(b, c) $\in$R, then (a, c) $\in$R

Here (1, 3) $\in$R and (3, 2) $\notin$R and (1, 2) $\in$ R

$\therefore$ R is not transitive

Hence, R are not reflexive, symmetric and transitive

The correct Answer is (d)

44. Consider the Boolean expression given by F =(X+Y+Z)(X'+Y)(Y'+Z), Which of the following Boolean expressions is/are equivalent to F (Complement of F)?

A. (X'+Y' + Z') (X+Y') (Y+Z')
B. (XY' + Z')
C. (X+Z') (Y'+Z')
D. (XY'+YZ' + X'Y'Z')

Choose the correct answer from the options given below.

(a) A, B and C only (b) B, C, and D only
(c) A, C and D only (d) A, B and D only

Ans. (b) : The given Boolean expression

F = (X + Y + Z) (X' +Y) (Y'+Z)

F' = [(X+Y+Z) (X' + Y)(Y'+Z)]'

using De Morgon's Law : (A+B)' = A'.B'

(A.B)' = A' + B'

F' = (X + Y + Z)' + (X' + Y)' + (Y' + Z)'

$\boxed{\text{F' = X'Y'Z' + XY' + YZ'}}$

F' = Y' (X+X'Z') + YZ'

[using property : A+A'B = (A+B)]

F' = XY' + Y'Z' + YZ' $\Rightarrow$ F' = XY' + Z' (Y'+Y)

$\boxed{\text{F' = XY' + Z'}}$

F' = Z' + XY'

F' = (Z' + X) (Z' + Y') [Using Distributive property A+BC = (A+B).(A+C)] So correct Answer is (b)

45. Which of the following statement is/are True?

A. The processor has direct access to both primary and secondary memory.

B. Primary memory stores the active instructions and data for the program being executed on the process

C. Secondary memory is used as a backup memory

D. Memory system is implemented on a single level memory

Chose the correct answer from the options given below :

(a) A and B only (b) B and C only

(c) A, B and C only (d) A and D only

Ans. (b) : The processor has direct access to both primary and secondary memory is not entirely accurate. While the processor can directly access primary memory (RAM), it typically does not have direct access to secondary memory.

■ Memory system is implemented on a single level memory is incorrect memory systems are typically implemented with multiple levels of memory hierarchy.

So correct Answer is option (b)

46. Inheritance provides the benefits of _______

A. shared interfaces

B. run-time polymorphism

C. less redundant information

D. dynamic binding

Choose the Correct answer from the options given below :

(a) A, B, C only (b) A, B, D only

(c) B, C, D only (d) A, C, D only

Ans. (a) : Inheritance provides the benefits of

■ Shared interface

■ Run time polymorphism

■ Less redundant information but dynamic binding achieved by polymorphism.

So correct Answer is (a)

47. Which of the following are correct using the constructor?

A. string s1;

B. string s2 (s1);

C. string s3{7};

D. string s4{'a'}

E. string s5{7, 'a'}

Choose the most appropriate answer from the options given below.

(a) A, D, E only (b) A, B, C only

(c) A, B, D only (d) A and B only

Ans. (d) : Statement A and B is correct but C, D and E is incorrect constructor does not use curly brackets.

So correct Answer is (d)

48. Suppose that a corporate website appearance varies according to the department that owns various documents. All the documents need to follow the corporate look and feel. The HR department does not need a separate style sheet. It needs only the sheet containing the difference from the corporate sheet.

Which of the following statements justify the above scenario?

A. The styles sheets cannot stack on each other

B. The style sheets can stack on each other.

C. The style sheets can override each other.

D. The style sheets cannot override each other.

Choose the most appropriate answer from the options given below :

(a) A, C only (b) B, C only

(c) B, D only (d) A, D only

Ans. (b) : The given statement

The HR department does not need a separate style sheet. When multiples style sheets are applied to a webpage, the rules defined in one style sheet can be overridden or modified by rule in another style sheet. This allows the HR department to have its own specific style sheet containing the differences from the corporate sheet. This rules in the HR style sheets will take precedence over the corresponding rules in the corporate style sheet, effectively customizing the appearance from that specific department while still maintaining the core corporate look and feel.

Therefore the correct statement is B and C.

So option (b) is correct.

49. Let Ti be a transaction that transfer Rs. 50 from account A to account B. This transaction can be defined as follows :

Step	Operation
1	Read (A)
2	A := A–50
3	Write (A)
4	Read (B)
5	B:=B+50
6	Write (B)

What will be the status of a database just after 3rd step but before 5th step of transaction execution?

(a) Always consistent

(b) May be temporarily inconsistent

(c) Must be abort

(d) Always rollback

Ans. (b) : The status of a data base use after 3^{rd} step but before 5^{th} step of transaction execution will be "May be temporarily inconsistent."

To understand why, let's look at what happens during the transaction execution :

1. Step 1 : Read (A) - The transaction reads the value of account A.
2. Step 2. A : = A–50 - The transaction deducts Rs. 50 from the value of A (let's say A was 100, So now A becomes 50).
3. Step 3: Write (A) - The transaction writes the new value of A(50) back to the database. If another transaction reads the value of A at this movement, it will see the intermediate state of A(50), which may not represent a consistent state of the database. This temporary inconsistency is resolved when the entire Ti transaction is completed (after executing step 6), and the database return to consistent state.

So option (b) is correct.

50. Consider the relation

Student (stuid, dept, clsroll)

The following functional dependencies are given :

FD1 : <u>stuid</u> → dept

FD2 : dept→stuid

FD3: stuid → clsroll

FD4: clsroll → stuid

FD5: dept → clsroll

Which functional dependencies should be removed to obtain the canonical cover of the set?

A. FD2 and FD3 B. FD1 and FD5
C. FD5 D. FD1

Choose the correct answer from the options given below :

(a) A, B only (b) A, C only
(c) B, C only (d) A, B, C only

Ans. (b) : The given relation
Student (<u>Stuid</u>, dept, clsroll)
The following function dependencies are given
FD1 : Stuid → dept
FD2 : dept → stuid
FD3 : stuid → clsroll
FD4 : clsroll → stuid
FD5 : dept → clsroll
So stuid is primary key. It can not removed.
So option (b) is correct.

51. Which of the following statements are true?

A. If the time quantum is extremely small, round robin can result in a large number of context switches.

B. The average turnaround time of a set of processes does not necessarily improve as the time-quantum size increases.

C. The average turnaround time can be improved if most processes finish their next CPU burst in a single time quantum.

Choose the correct answer from the options given below :

(a) A only (b) A, B only (c) B, C only (d) A, B, C

Ans. (d) :
A. If the time quantum is extremely small, round robin can result in a large number of context switches.
B. The average turnaround time of a set of processes does not necessarily improves the time - quantum size increases.
C. The average turnaround time can be improved if most processes finish their next CPU burst in a single time quantum.
So All statement is true.

52. Consider the following set of processes that need to be scheduled on a single CPU. All the times are given in milliseconds.

Using the shortest remaining time first scheduling algorithm, the average process turnaround (in milliseconds) is _______

(a) 7.2 (b) 8.2 (c) 6 (d) 9

Ans. (a) :

Process Name	Arrival Time	Execution time
A	0	6
B	3	2
C	5	4
D	7	6
E	10	3

CPU scheduled

A	B	A	C	E	D

0 3 5 8 12 15 21

Average Turnaround Time :

$$=\frac{(8-0)+(5-3)+(12-5)+(21-7)+(15-10)}{5}$$

$$=\frac{8+2+7+14+5}{5}=\frac{36}{5}$$

Average Turnaround Time = 7.2
So option (a) is correct.

53. Select the correct order of basic page replacement :

A. page replacement algorithm is used in case there is no free frame.

B. The process is continued by restarting the instruction caused by TRAP

C. Location of desired frame is found from disk.

D. Desired page is brought into newly freed frame and page & frame tables are updated

(a) A, B, D, C (b) C, D, A, B

(c) C, A, D, B (d) A, D, B, C

Ans. (c) :

C. Location of desired frame is found from disk

A. Page replacement algorithm is used in case there is no free frame.

D. Desired page is brought into newly freed frame and page & frame tables are updated.

B. The process is continued by restarting the instruction caused by TRAP

So correct order of basic page replacement is

C, A, D, and B

So option (c) is correct Answer.

54. Given below are two statements :

Statement I : Software testing must be recognized as a significant effort, and its planning must begin at the earliest possible time.

Statement II : The cost of software testing is an important contribution in determining overall software development costs. In the light of the above statements, choose the correct answer from the options given below.

(a) Both Statement I and Statement II are true

(b) Both Statement I and Statement II are false.

(c) Statement I is true but Statement II is false.

(d) Statement I is false but Statement II is ture.

Ans. (a) : Both statement are true.

Explanation :

1. The first statement emphasizes that software testing should be considered a significant effort and should be planed from the earliest stages of software development. This is a valid point as early planning for testing ensures that potential issues can be identified and resolved early on, which can save time and resources in the long run.
2. The second statement states that cost of software testing is a crucial factor in determining the overall software development costs. This is also true because testing involves various activities such as designing test cases executing tests, debugging, and fixing defects, all of which require resources and effort.

In conclusion, both statement are valid and emphasize the importance of software testing in the software development process.

55. Which of the following need not be assessed during unit testing?

A. Algorithmic performance

B. Number of lines of code.

C. Error handling.

D. Execution paths.

Choose the correct answer from the options given below :

(a) A, B, C only (b) A, B, D only

(c) A, C, D only (d) B, C, D only

Ans. (d) : The correct Answer is (d) B, C, D only
Let's go through each option to understand why.

A. Algorithmic performance : This is typically not assessed during unit testing. Unit testing is concerned with verifying the correctness of the unit's functionality, not its performance characteristics

B. Number of line of code : The number of line of code is not relevant to unit testing. Unit tests are designed to check behavior of the unit's functionality, regardless of how many lines of code are used to implement it.

C. Error handling : Error handling is an essential aspect of unit testing. During unit testing, you want ot ensure that the unit handles errors and exceptions properly and produces the expected results in such scenarios.

D. Execution paths : Unit testing aims to test aims to test different execution path of the unit's code to ensure all possible scenarios are covered. It helps in identifying and fixing bugs related to different path through the unit code.

56. Which of the following rotations are correctly explained in the context of an AVL tree.

A. LL rotation : The new node is insterted in the left sub-tree of the left sub-tree of the critical node.

B. RR rotation : The new node is inserted in the right sub-tree of the right sub-tree of the critical node.

C. LR rotation : The new node is inserted in the right sub-tree of the left sub-tree of the critical node.

D. RL rotation : The new node is inserted in the left sub-tree of the right sub-tree of the critical node

Choose the correct answer from the options given below.

(a) A, D only (b) A, C only
(c) A, B only (d) B, C only

Ans. (*) : An AVL tree defined as a self - balancing Binary search Tree (BST). Where the difference between heights of left and right sub trees for any node cannot be more than one.
Rotating the sub trees in an AVL tree.

■ Left Rotation
■ Right Rotation
■ Left-Right Rotation
■ Right-left Rotation

UGC has drop this question.

57. In the context of a Hamiltonian problem which of the following are correct?

A. Hamiltonian cycle problem is NP-complete.

B. If G is an undirected Bipartite graph with an odd number of vertices, then G is Hamiltonian.

C. Hamiltonian path problem cannot be solved in polynomial time on directed acylic graph.

D. If G is a connected undirected graph with at least three vertices and G^3 is the graph obtained by connecting all pairs of vertices that are connected by a path in G of length at most 3.

E. If any NP complete problem is polynomial time solvable the P = NP.

Choose the most appropriate answer from the options given below.

(a) A, B, C only (b) B, D, E only
(c) A, D, E only (d) A, C, E only

Ans. (d) : Let's go through each option to explain why
A. Hamiltonian problem is NP=complete : The Hamiltonian path problem is known to be NP-complete, which means that its is unlikely to have a polynomial-time algorithm to solve it and it is at least as hard as any problem in the complexity class NP.
B. This statement is false. A Hamiltonian cycle (or path) in a graph with an old number of vertices is not always guaranteed.
C. This statement is true. The Hamiltonian path problem is NP-Complete for both directed and undirected graphs. A directed a cyclic graph (DAG) is a spcial case where there cannot be any Hamiltonian path, as it would require visiting each vertex exactly once without forming any cylces which is not possible in a DAG.
D. This statement is false. The notation G^3 typically refers to the cube of a graph and it does not relate to finding Hamiltonian path or cycles.
E. If any NP complete problem is polynomial time solvable the P = NP
This statement is true.
Therefore, the correct statements are A, C and E.

58. For matrix multiplication, if we use divide and conquer algorithm, then _____
A. the running time is $T(n) = \Theta(n^3)$
B. the running time is $T(n) = \Theta(n^2 \log n)$
C. The recurrence can be defined as $T(n) = 8T(n/2) + \Theta(n^2)$
D. the recurrence relation is $T(n) = 8T(2n) + \Theta(n^2 \log n)$
E. Masters theorem, one of the cases is applicable
Choose the most appropriate answer from the options given below.
(a) A, C, E only (b) A, D, E only
(c) B, D, E only (d) B, C, D only

Ans. (a) : Matrix multiplication, if we use decide and conquer algorithm, them.
A. The running time is $T(n) = \theta(n^2)$
This statement is true.
B. The recurrence can be defined a
$T(n) = 8T(n/2) + \theta(n^2)$
When using the decide and conquer algorithm for multiplication, the recurrence relation is typically of the form.
$T(n) = aT\left(\frac{n}{b}\right) + f(n)$, where "a" represents the number of sub problems, "b" is the size of each sub problem, and "f(n)" denote the time completely for combining the sub problem solution.
This indicates that the running time of the algorithm is $\theta(n^2)$ since each level of the recursion involves $\theta(n^2)$ work and there are log(n) level in total.
So option (a) is correct.

59. Which of the following Finite Automata, recognizes the language L=(a, b)*?
A. M =({q0}, (a,b},δ,q0{q0)}) where δ is

	a	b
q0	{q0}	{q0}

B. M =({q0, q1}, {a,b}, δ, q0,{q0, q1}) where δ is

	a	b
q0	{q0,q1}	{q0,q1}
q1	{q1}	{q1}

C. M =({q1, q1}, {a,b}, δ, q0,{q0, q1}) where δ is

	a	b
q0	{q0, q1}	{q0}
q1	-	{q1}

D. M =({q0, q1,q2}, {a,b}, δ, q0,{q0}) where δ is

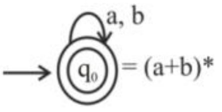
	a	b
q0	{q0}	{q0}
q1	{q2}	-
q2	{q0}	{q2}

(a) A and B (b) A and D
(c) A, B and C (d) A, B, C and D

Ans. (d) : The given language
L = {a, b}*
q_0 = initial state, input = a, b, δ–Transition state $\{q_0\}$ final state.

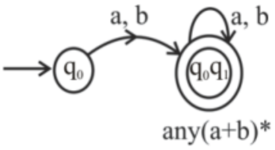
A.

	a	b
q_0	$\{q_0\}$	$\{q_0\}$



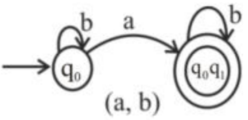
B.

	a	b
q_0	$\{q_0, q_1\}$	$\{q_0, q_1\}$
q_1	$\{q_1\}$	$\{q_1\}$



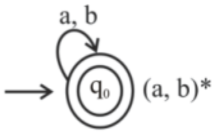
C.

	a	b
q_0	$\{q_0, q_1\}$	$\{q_0\}$
q_1		$\{q_1\}$



D.

	a	b
q_0	$\{q_0\}$	$\{q_0\}$
q_1	$\{q_2\}$	-
q_2	$\{q_0\}$	$\{q_2\}$



So option (d) is correct

60. Which of the following statement(s) are correct?
A. Syntax Directed Translation can be applied in Type Checking.
B. Syntax Directed Translation can be used in 3-address code generation.
C. Syntax Directed Translation can be used in construction of syntax tree.
D. Syntax Directed Translation can be used to recognize valid tokens.
Choose the most appropriate answer from the options given below :
(a) A, B and C (b) A, B, C and D
(c) A, C and D (d) B, C and D

Ans. (a) : Statement A, B and C is correct but D is false.
Syntax directed translation can be used to recognize valid token.
Correct - Syntax Directed Translation can be used to define the translation rules for recognizing and generating valid tokens during lexical analysis.

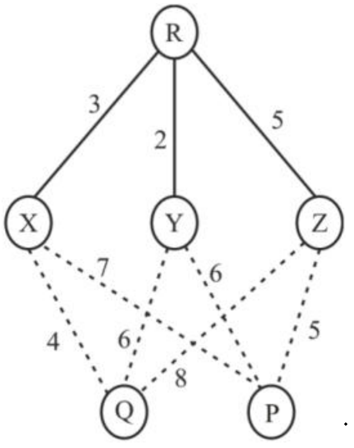
61. **Consider a computer network using the distance vector routing algorithm in its network layer. The objective is to find the shortest cost path from router R to routers P and Q. Assume that R does not initally know the shrotest routes to P and Q. Assume that R has three neighboring routers, namely; X, Y and Z. During one iteration R measures its distance as 3, 2 and 5, 7, 6 and 5 from router X, Y and Z to router P, respectively. The routing vector also indicates that the distance to router Q from Routers X, Y and Z are 4, 6, 8 respectvely. Which of the following statement is/are True with respect to the new routing table at router R after updation during this iteration.**

A. The distance from R to P will be stored as 10.
B. The distance from R to Q will be stored as 7.
C. The next hop router for a packet from R to P is Y.
D. The next hop router for a packet from R to Q is Z

Choose the correct answer from the option given below.

(a) B and C only (b) A and C only
(c) C and D only (d) B and D only

Ans. (a) :



As per Distance Vector routing Algorithm, R to
P = min {R – X – P, R–Y–P, R–Z–P}
= {10, 8, 10} = 8
R to Q = min{R – X – Q, R – Y – Q, R – Z – Q}
= min {7, 8, 13} = 7
Hence statements (B) and (C) are correct.
So option (a) is correct.

62. **Consider the routing table of an organization's router as shown in the table given below**

Subnet Number	Subnet Mask	Next Hop
12.20.164.0	255.255.252.0	R1
12.20.170.0	255.255.254.0	R2
12.20.168.00	255.255.254.0	Interface 0
12.20.166.0	255.255.254.0	Interface 1
Default		R3

Which of the following prefixes in CIDR Notation can be collectively used to correctly aggregate all of the subnets in the routing table?

A. 12.20.164.0/20 **B. 12.20164.0/22**
C. 12.20.164.0/21 **D. 12.20.168.0/22**

Choose the correct answer from the options given below :

(a) A and B only (b) A and C only
(c) C and D only (d) B and D only

Ans. (d) : The given subnet number
1. 12.20.164.0 = 12.20.10100100.0
2. 12.20.170.0= 12.20 10101010.0
3. 12.20.168.0 = 12.20.10101000.0
4. 12.20.166.0 = 12.20.10100110.0

1 and 4 first 6 bits are same.
So merging
12.20.101001001/22
12.20.164.0/22
So statement (B) is correct
again 2 and 3 first 6 bits are same so merging.
12.20.10101000.0/22
12.20.168.0/22
So statement (D) is correct.
Hence option (d) is correct.

63. Consider two hosts P and Q are connected through a router R. The maximum transfer unit (MTU) value of the link between P and R is 1500 bytes and between R and Q is 820 bytes. A TCP segment of size of 1400 bytes is transferred from P to Q through R with IP identification value of 0x12334. Assume that IP header size is 20 bytes. Further the packet is allowed to be fragmented that is Don't Fragment (DF) flag in the IP Header is not set by P.

Which of the following statement is/are not true?

A. Two fragments are created at R and IP datagram size carrying the second fragment is 620 bytes.

B. If the second fragment is lost then R resend the fragment with IP Identification value of 0x1234

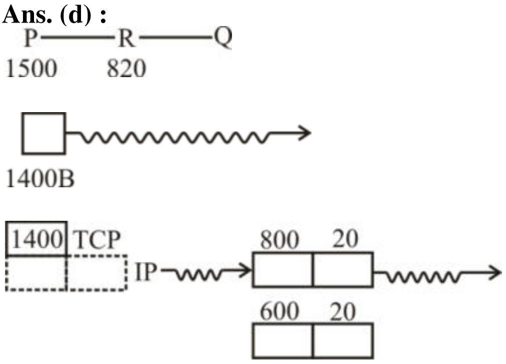
C. If the second fragment lost then P required to resend the entire TCP segment.

D. TCP destination port can be determined by analyzing the second fragment only.

Choose the correct answer from the options given below.

(a) A and B only (b) A and C only
(c) C and D only (d) B and D only

Ans. (d) :



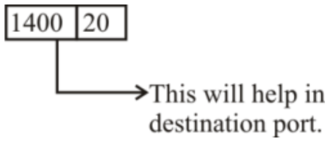
2. fragments
Now if 2[nd] fragment is lost, as it is TCP, (Connection oriented) So P will send again the whole data (1400) → within MTV of P → R link.
Now its R who has the job of fragmentation, so R will send both packets again.
A, C are true

B, D are false

B is false as ID number changes on re-transmission.

D is false as at destination, reassembly of fragments is done.

So



So option (d) is correct.

64. Consider the following features of Neural networks.

A. It is essentially a machine learning algorithm.

B. It is useful when solving problems for which the data set is very large.

C. They are able to extract features without input from the programmer.

D. These are systems modeled on the human brain and nervous system.

Choose the most appropriate answer from the options given below.

(a) A only (b) A & B only

(c) A, B & C only (d) A, B, C & D

Ans. (d) : Let's consider the features of neural networks you mentioned :

A. Correct, Neural networks are indeed a machine learning algorithm commonly used for various tasks like image recognition natural language processing and more.

B. Correct, Neural networks can handle large datasets effectively making them suitable for complex problems with vast amounts of data.

C. Correct, That's one of the key advantages of neural network-they can automatically learn and extract relevant features from the input data without explicit programming.

D. Correct, Neural network are inspired by the structure and function of the human brain and nervous system. They consist of interconnected artificial neurons that process and transmit information mimicking biological neural networks.

So option (d) is correct.

65. Consider the following draw backs is/are of Rule based approach is/are :

A. The learning is static.

B. Any changes made to the original data will not be considered.

C. Once trained the model cannot improve on the basis of feedback.

Choose the correct answer from the options given below.

(a) All the above statements are correct

(b) The statements A & B only are not correct

(c) The statements B & C only are not correct

(d) All the above statements are not correct

Ans. (a) : The correct statement is A, B, and C the draw pack of the Rule - based approach are

A. The learning is static

B. Any changes made to the original data will not be considered.

C. Once trained the model cannot improve on the basis of feedback.

These limitations highlight the inflexibility and lack of adaptability of Rule- based system compared to other machine learning approaches.

66. Match List I with List II

List-I		List-II	
A.	**Linear Program**	**I.**	**Prim's Algorithm**
B.	**Predate Logic**	**II.**	**In order**
C.	**Tree Traversal**	**III.**	**Simplex Algorithm**
D.	**Minimum Spanning Tree Problem**	**IV.**	**First order logic**

Choose the correct answer from the options given below :

(a) A-IV; B-III; C-I; D-II
(b) A-III; B-IV; C-II; D-I
(c) A-II; B-IV; C-I; D-III
(d) A-III; B-IV; C-I; D-II

Ans. (b) : The Correct is

A.	Linear Program	III.	Simplex Algorithm
B.	Predate Logic	IV.	First order logic
C.	Tree Traversal	II.	In order
D.	Minimum Spanning Tree Problem	I.	Prim's Algorithm

So option (b) is correct.

67. Match List I with List II

List-I		List-II	
A.	**DDA**	**I.**	**Program Control instruction**
B.	**LXI**	**II.**	**Interrupt instruction**
C.	**RST**	**III.**	**Data Movement instruction**
D.	**JMP**	**IV.**	**Athematic instruction**

Choose the correct answer from the options given below :

(a) A-IV; B-II; C-I; D-III
(b) A-IV; B-III; C-II; D-I
(c) A-IV; B-II; C-III; D-I
(d) A-III; B-IV; C-II; D-I

Ans. (b) : The Correct is

A.	DDA	IV.	Athematic instruction
B.	LXI	III.	Data Movement instruction
C.	RST	II.	Interrupt instruction
D.	JMP	I.	Program Control instruction

So correct answer is option (b)

68. Match List I with List II

List-I		List-II	
A.	**2-D Array Input**	**I.**	**Liquid Crystal display**
B.	**Transmissive device**	**II.**	**Flat-bed scanner**
C.	**Binary image**	**III.**	**Camera**
D.	**1-D array input**	**IV.**	**Ink jet printer**

Choose the correct answer from the options given below :

(a) A-I; B-III; C-II; D-IV
(b) A-III; B-I; C-IV; D-II
(c) A-III; B-I; C-II; D-IV
(d) A-I; B-II; C-IV; D-III

Ans. (b) : The Correct is			
A.	2-D Array Input	III.	Camera
B.	Transmissive device	I.	Liquid Crystal display
C.	Binary image	IV.	Ink jet printer
D.	1-D array input	II.	Flat-bed scanner
So correct Answer is Option (b)			

69. Match List I with List II

List-I		**List-II**	
A.	**Timestamp protocol**	**I.**	**As long as the transaction is already running, it will try to obtain the required lock in the first part**
B.	**Strict two-phase locking protocol**	**II.**	**Indicates final lock of the schedule**
C.	**Two-phase locking**	**III.**	**Ensures freedom from deadlock**
D.	**Lock point in two phase locking protocol**	**IV.**	**Ensures cascade-less rollback**

Choose the correct answer from the options given below :

(a) A-III; B-IV; C-II; D-I
(b) A-III; B-IV; C-I; D-II
(c) A-IV; B-III; C-II; D-I
(d) A-IV; B-II; C-I; D-III

Ans. (b) : The Correct is			
A.	Timestamp protocol	III.	Ensures freedom from deadlock
B.	Strict two-phase locking protocol	IV.	Ensures cascade-less rollback
C.	Two-phase locking	I.	As long as the transaction is already running, it will try to obtain the required lock in the first part
D.	Lock point in two phase locking protocol	II.	Indicates final lock of the schedule
So option (b) is correct.			

70. Match List I with List II

List-I		**List-II**	
A.	**(reference bit=0, modify bit=0)**	**I.**	**Must write out before replacement**
B.	**(reference bit=0, modify bit=1)**	**II.**	**Probably will be used again soon and need to write out.**
C.	**(reference bit=1, modify bit=0)**	**III.**	**Best page to replace**
D.	**(reference bit=1, modify bit=1)**	**IV.**	**Probably will be used again soon**

Choose the correct answer from the options given below :

(a) A-II; B-I; C-III; D-IV
(b) A-II; B-I; C-IV; D-III
(c) A-III; B-I; C-II; D-IV
(d) A-III; B-I; C-IV; D-II

Ans. (d) : The Correct is			
A.	(reference bit=0, modify bit=0)	III.	Best page to replace
B.	(reference bit=0, modify bit=1)	I.	Must write out before replacement
C.	(reference bit=1, modify bit=0)	IV.	Probably will be used again soon
D.	(reference bit=1, modify bit=1)	II.	Probably will be used again soon and need to write out.
So option (d) is correct.			

71. Match List I with List II

List-I		**List-II**	
A.	**Condition coverage**	**I.**	**Black box testing**
B.	**Equivalence class partitioning.**	**II.**	**System testing**
C.	**Volume testing**	**III.**	**White box testing**
D.	**Alpha testing**	**IV.**	**Performance testing**

Choose the correct answer from the options given below :

(a) A-II; B-III; C-IV; D-I
(b) A-III; B-I; C-IV; D-II
(c) A-IV; B-I; C-II; D-III
(d) A-III; B-IV; C-I; D-II

Ans. (b) : The Correct is			
A.	Condition coverage	III.	White box testing
B.	Equivalence class partitioning.	I.	Black box testing
C.	Volume testing	IV.	Performance testing
D.	Alpha testing	II.	System testing

72. Match List I with List II

List-I		**List-II**	
A.	**8-Queens problem**	**I.**	**Divide and conquer**
B.	**Closest pair of Points**	**II.**	**Dynamic Programming**
C.	**String Edit problem**	**III.**	**Greedy Method**
D.	**Knapsack problem**	**IV.**	**Backtracking**

Choose the correct answer from the options given below :

(a) A-I; B-II; C-III; D-IV
(b) A-I; B-IV; C-II; D-III
(c) A-IV; B-I; C-II; D-III
(d) A-II; B-III; C-IV; D-I

Ans. (c) : Correct Match are -			
A.	8-Queens problem	IV.	Backtracking
B.	Closest pair of Points	I.	Divide and conquer
C.	String Edit problem	II.	Dynamic Programming
D.	Knapsack problem	III.	Greedy Method

73. Match List I with List II

	List-I		**List-II**
A.	**Initial Configuration of stack and input in LL Parsing process.**	**I.**	**\$ w\$**
B.	**Final Configuration of stack and input in LL Parsing process**	**II.**	**\$ \$**
C.	**Initial Configuration of stack and input in LR Parsing process.**	**III.**	**\$S w\$**
D.	**Final Configuration of stack and input in LR Parsing process**	**IV.**	**\$S \$**

Choose the correct answer from the options given below :

(a) A-III; B-IV; C-I; D-II
(b) A-III; B-II; C-I; D-IV
(c) A-II; B-III; C-IV; D-I
(d) A-II; B-IV; C-I; D-III

Ans. (b) : Correct Match are -			
A.	Initial Configuration of stack and input in LL Parsing process.	III.	\$S, w\$
B.	Final Configuration of stack and input in LL Parsing process	II.	\$, \$
C.	Initial Configuration of stack and input in LR Parsing process.	I.	\$ w\$
D.	Final Configuration of stack and input in LR Parsing process	IV.	\$S \$

74. Match List I with List II

	List-I		**List-II**
A.	**Physical layer**	**I.**	**Transforming the raw bits in the form of frame for transmission.**
B.	**Data Link Layer**	**II.**	**Control and monitoring of subnet.**
C.	**Network Layer**	**III.**	**Transmission of raw bits over communication channel**
D.	**Transport layer**	**IV.**	**Datagrams Transmission data through connection oriented or connection less using datagrams.**

Choose the correct answer from the options given below :

(a) A-III; B-II; C-I; D-IV
(b) A-II; B-III; C-I; D-IV
(c) A-III; B-I; C-II; D-IV
(d) A-II; B-IV; C-I; D-III

Ans. (c) : The Correct is			
A.	Physical layer	III.	Transmission of raw bits over communication channel
B.	Data Link Layer	I.	Transforming the raw bits in the form of frame for transmission.
C.	Network Layer	II.	Control and monitoring of subnet.
D.	Transport layer	IV.	Datagrams Transmission data through connection oriented or connection less using datagrams.

75. Match List I with List II

List-I		List-II	
A.	**Regression**	**I.**	**Unsupervised Learning**
B.	**KNN**	**II.**	**Supervised Learning**
C.	**Clustering**	**III.**	
D.	**Decision Tree**	**IV.**	

Choose the correct answer from the options given below :

(a) A-I; B-II; C-I; D-II
(b) A-II; B-II; C-I; D-II
(c) A-I; B-II; C-II; D-I
(d) A-II; B-II; C-II; D-I

Ans. (b) : The Correct is			
A.	Regression	II.	Supervised Learning
B.	KNN	II.	Supervised Learning
C.	Clustering	I.	Unsupervised Learning
D.	Decision Tree	II.	Supervised Learning
So option (b) is correct.			

76. Let G be a simple undirected graph with n(n>0) vertices and 'k' connected components. What is the minimum possible number of edges of 'G'?

(a) k(n–1) (b) n(k–1)
(c) n+k–2 (d) n–k

Ans. (d) : The minimum possible number of edges in a simple undirected graph G with n vertices and 'k' connected components is n-k.
So correct Answer is option (d)

77. The sequence of events that happen during a typical fetch operation is/are :

A. Memory **B. MBR** **C. MAR**
D. PC **E. IR**

Choose the correct order from the options given below :

(a) A, B, C, D, E (b) D, C, A, B, E
(c) E, D, C, B, A (d) D, C, A, E, B

Ans. (b) : During a typical fetch operation the sequence of events is a follows.

- The program counter (PC) holds the address of the next instruction to fetched.
- The Memory address resister (MAR) receives the address from the PC, indicating the location of the in memory.
- The memory unit retrieves the instruction from the memory location specified by the MAR.
- The fetched instruction is stored in the memory Buffer register.
- The fetched instruction is also loaded into the instruction register (IR) allowing the processor to decode and execute the instruction.

So option (b) is correct.

78. Arrange the following search and sort algorithms in ascending order of their worst-case complexities.

A. Merge sort **B. Linear Search**

C. Bubble Sort **D. Binary Search**

Choose the correct answer from the options given below :

(a) A, B, D, C (b) B, D, A , C

(c) B, D, C, A (d) B, A, D, C

Ans. (b) : Here are the search and sort algorithms arranged in ascending order of their worst-case complexities.

B. Linear search - worst-cost complexity : O(n)

D. Binary search - worst - case complexity : O(log n)

A. Merge sort - worst - case complexity : O (log n)

C. Bubble sort - worst - case complexity : O (n^2)

So option (b) is correct.

79. Consider the following schedule S.

T1	T2	T3	T4	Timestamp
	R(X)	R(Y)		t1
	W(X)	W(Y)		t2
R(Y)			R(X)	t3
W(Y)			W(X)	t4
		R(X)		t5
		W(X)		t6

R(X) denotes read operation on data item X by transaction Ti.

W(X) denotes write operation on data item X by transaction Ti.

Choose the correct execution order for the above schedule S.

(a) T3-T2-T1-T4 (b) T2-T3-T1-T4

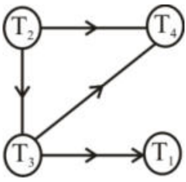
(c) T2-T4-T3-T1 (d) T4-T2-T3-T1

Ans. (c) : The given schedule S.

T1	T2	T3	T4	Timestamp
	R(X)	R(Y)		t1
	W(X)	W(Y)		t2
R(Y)			R(X)	t3
W(Y)			W(X)	t4
		R(X)		t5
		W(X)		t6

R(X) denotes read operation on data item X by transaction Ti

W(X) denotes write operation on data item X by transaction Ti



So T2 should be start

T1 should be end

T2 – T4 – T3 – T1

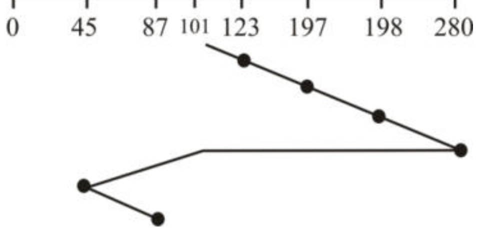
So option (c) is correct.

80. A disk has 300 cylinders (0 to 299). If the initial position of the read-write head is at cylinder 101, moving towards the higher end of the disk calculate the total head movement following C_LOOK policy for the read request sequence. 123, 45, 197, 87, 250, 198, 280

(a) 456 (b) 450

(c) 458 (d) 455

Ans. (a) :



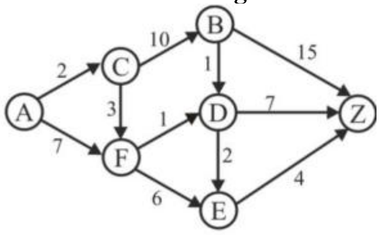
The total head movement the read request sequence.
(280-101) + (280-45) + (87-45) = 456
So correct Answer is option (a).

81. The incremental model of software development is ______

(a) A reasonable approach when requirements are well defined.
(b) A good approach when a working core product is required quickly.
(c) The best approach to use for project with large developments teams.
(d) A revolutioary model that is not used for commerical products.

Ans. (b) : The incremental model of software development is a good approach when a working core product is required quickly.

82. Consider the graph given below. What is the order in which the nodes mentioned in the options are to be traversed to find the shortest path from A to Z using best first search?

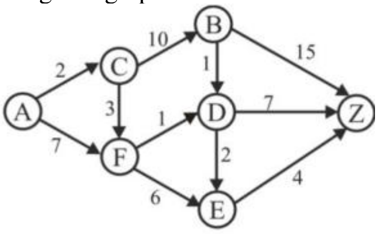


A. Node-A **B. Node-B**
C. Node-C **D. Node-D**
E. Node-E **F. Node-F**

Choose the correct answer from the options given below

(a) A→C→F→D→E (b) A→C→E→B
(c) A→C→F→E→B (d) A→C→D→F

Ans. (a) : The given graph



By given option
A → C → F → D → E
So option (a) is correct.

83. The precedence of regular expression operators from lowest to highest precedence is :

(a) Star, Concatenation, Union
(b) Union, Star, Concatenation
(c) Union, Concatenation, Star
(d) Star, Union, Concatenation

Ans. (c) : The precedence of regular expression operators from lowest to highest precedence is -
Star is highest then concatenation and union is lowest precedence.
So option (c) is correct.

84. Arrange the following in the correct order which respect to key generation for RSA algorithm.

A. Choose an integer e such that 1 < e < λ(n)

B. Compute Key length n = p*q

C. Find d which is equal to 1/e

E. Compute λ(n)

Choose the correct answer from the options given below :

(a) A-B-C-D-E (b) D-C-B-E-A
(c) D-B-E-A-C (d) A-B-D-E-C

Ans. (c) : Here is the correct order for key generation in the RSA algorithm :

- Choose two large prime numbers P and Q.
- Compute the key length n = p*q.
- Compute λ(n) (also know as the carmichael's totient function for RSA) which is the least common multiple of p–1 and q–1.
- Choose an integer e such that 1 < e < λ(n) and e is coprime with λ(n)
- Find the value of d, which is the multiplication inverse of e module λ(n). This means finding d such that (*e)%λ(n) = 1

After following these steps you will have the public key (e, n) and the private key (d, n) for the RSA algorithm.

85. Which of the following is related to function of layers in a neural network?

A. Information - Input layer - Hidden layer

B. Blas - Output layer - hidden layer.

C. Hidden Layer - alogrithm - Output layer

D. Input layer - output layer - algorithms

Choose the correct answer from the options given below.

(a) Only sequence 'A' is correct.
(b) Only sequence 'B' is correct
(c) Only sequence 'C' is correct
(d) Only sequence 'D' is correct

Ans. (c) : The correct option related to the function of layers in a neural network is
C. Hidden - Algorithm - Output layer

86. Given below are the two statements:

Statement I : The main objective for using virtual memory is to increase the effective capacity of the memory system.

Statement II : The size of virtual memory can be larger than the main memory.

In the light of the above statements, choose the correct answer from the options given below.

(a) Both Statement I and Statement II are true.
(b) Both Statement I and Statement II are false
(c) Statement I is true but Statement II is false
(d) Statement I is false but Statement II is true.

Ans. (a) : The given statement.
Statement I : The main objective for using virtual memory is to increase the effective capacity of the memory system. This statement is true.
Statement II : The size of virtual memory can be larger than main memory. It is also true.
So option (a) is correct.

87. Given below are the two statements :

Statement I : Any retireval request that is specified in the basic relational algebra can also be specified in relational calculus.

Statement II : Any retrieval request that is specified in the relational calculus can also be specified in basic relational algebra.

In the light of the above statements, choose the correct answer from the options given below.

(a) Both Statement I and Statement II are true.
(b) Both Statement I and Statement II are false
(c) Statement I is true but Statement II is false
(d) Statement I is false but Statement II is true.

Ans. (a) : The given statement
Statement I : Any retrieval request that is specified in the basic relational algebra can also be specified in relational calculus. This is true.
Statement -II : Any retrieval request that is specified in the relational calculus an also be specified in basic relational algebra.
This statement is true.
So option (a) is correct.

88. The primary objectives of the SWE-IPT are to :

A. Solicit stake holder needs and expectations.
B. Specify the software requirements for the product and post-development processes.
C. Generate the integrated technical plans for accomplishing the software development effort.
D. Translate stakeholder requirements into a complete, specified architectural description.
E. Ensure that the software technical data package is sufficiently detailed to facilitate an efficient and effective software implementation.

Choose the correct answer from the options given below :

(a) All the statements are correct
(b) Only B & C are correct
(c) Only A & E are correct
(d) Only C, D & E are correct

Ans. (a) : You are taking about the software Engineering Integrated Product Team (SWE - IPT) objectives.
The primary objectives of the SWE - IPT are as follows:

1. Solicit stakeholder needs and expectations.
2. Specify the software requirements for the product and post-development processes.
3. Generate the integrated technical plans for accomplishing the software development effort.
4. Translate stakeholder requirements into a complete, specified architectural description.
5. Ensure that the software technical data package is sufficiently detailed to facilitate an efficient and effective software development process.

So option (a) is correct.

89. Consider the following statements:

A. For every regular language, we can design Turing Machine.
B. For every context free language, we can design Turing Machine.

Which of the following is correct?

(a) Statement-A is false and Statement-B is true.
(b) Statement-A and Statement-B both are false.
(c) Statement-A is true and Statement-B is false
(d) Statement-A and Statement-B both are true

Ans. (d) : Both statement (A) and (B) are correct.

A. For every regular language, we can design a Turing Machine. Regular language are equivalent in power to the languages that can be recognized by a Turing Machine, So there exists a Turing Machine for every regular language.

B. For every context free language, we can design a Turing Machine. Similarly, context free language are also Turing -recognizable meaning there exists a Turing Machine that can recognize every context-free language.

90. Given below are two statements one is labelled as Assertion A and the other is labelled as Reason R

A. Artificial Neural Networks are the backbone of deep learning algorithms.

R. Back propagation is one of the techniques used to train an ANN.

In the light of the above statements, choose the most appropriate answer from the options given below :

(a) Both A and R are correct and R is correct explanation of A.
(b) Both A and R are correct but R is NOT the correct explanation of A
(c) A is correct but R is not correct
(d) A is not correct but R is correct

Ans. (b) : The most appropriate answer is Both A and R are correct but R is Not the correct explanation of A.

Explanation

Artificial Neural network are the backbone of deep learning algorithms.

Back propagation or backward propagation of error is an algorithm that is designed to test for errors working back from output nodes to input nodes. It is an important mathematical tool for improving the accuracy of predications in data mining and machine learning. Essentially, back propagation is an algorithm used to calculate derivation quickly.

So option (b) is correct.

Use the following schema of the academic institution relational database :

Student (<u>rollno</u>, name, degree, year, sex, deptno, advisor)

department (<u>deptid</u>, name, hod, phone)

professor (<u>empid</u>, name, sex, startyear, deptno. phone)

course (<u>courseid</u>, cname, creidts, deptno)

enrollment (<u>rollno</u>, <u>courseid</u>, <u>sem</u>, <u>year</u>, grade)

teaching (<u>empid</u>, <u>courseid</u>, <u>sem</u>, <u>year</u>, classroom)

prerequisite (<u>precourseid</u>, <u>courseid</u>)

deptno is foreign key in the student, professor and course relations referring to deptid of department relation;

advisor is a foreign key in the student relation referring to empid of professor relation;

hod is a foreign key in the department relation referring to empid of professor relation;

rollno is a foreign key in the enrollment relation referring to rollno of student relation;

coursied is a foriegn key in the enrollment, teaching relations referring to courseid of course relation;

empid is a foreign key of the teaching relation referring to empid of professor relation;

precouseid and courseid are foreign keys in the presquisite relation referring to courseid of the course relation; (91-95).

91. Which of the following queries would retrieve the roll number and names of students who have enrolled for all the prerequisite course of the course number 324?

(a) Select s.rollno, s.name From student s, enrollment e Where e.rollno = s.rollno and e.courseid = ANY(select precouseied from prerequisite p where p.courseid = "324")
(b) Select s.rollno, s.name From student s, enrollment e Where e.rollno = s.rollno and e.courseid = ALL(select precouseied from prerequisite p where p.courseid = "324")
(c) Select s.rollno, s.name From student s Where NOT EXISTS (select * prerequisite p where p.courseid = "324") and NOT EXISTS (select * from enrollment e where e.courseid = p.precoursied and e.rollno = s.rollno.)
(d) Select s.rollno, s.name From student s Where EXISTS (select * prerequisite p where p.courseid = "324") and EXISTS (select * from enrollment e where e.courseid = p.precoursied and e.rollno = s.rollno.)

Ans. (c) : The correct query to retrieve the roll number and name of students who have enrolled for all the prerequisite courses of the course number 324.

C. Select s.rollno, s.name from student s Where NOT EXISTS (select * prerequisite p where p.courseid = "324") and NOT EXISTS (select * from enrollment e where e.courseid = p.precoursied and e.rollno = s.rollno.)

92. Given query : select s.rollno from students where s.sex = ALL (select f.sex from professor f where f.empid = s.advisor)

Consider the following statements about the given query that retrieves the set of all students whose gender is same as the gender of their advisor:

S1. It has a correlated subquery

S2. It has an uncorrelated subquery

S3. If we replace ALL by ANY in the above query, its result will be different.

S4. If we replace ALL by ANY in the above query, its result will be same.

Identify all the statement that are TRUE :

(a) S1 and S3 are true (b) S1 and S4 are true
(c) S2 and S3 are true (d) S2 and S4 are true

Ans. (b) : The correct query to retrieves the set of all students whose gender is same as the gender of their advisor :

S1. It has a correlated subquery

S4. If we replace ALL by ANY in the above query, its result will be same.

So option (b) is correct.

93. Which of the following queries would compute, for each department, the total credits of all the courses offered by the department?

(a) select deptid, name, sum(credits) as totalcredits from department, course where deptid = deptno group by deptid, name;
(b) select depid, name, count(credits) as totalcredits from department, course where deptid=deptno group by deptid, name;
(c) select deptid, name, sum(credits) as totalcredits from department, course where deptid = deptno
(d) select depid, name, sum(credits) as total credits from department, course group by deptid, name having deptid = deptno;

Ans. (a) : The correct query to compute for each department, the total credits of all the courses offered by the department -
Select deptid, name, sum(credits) as totalcredits from department, course where deptid = deptno.

94. Which of the following queries would retrieve the dept id and name of the all departments that are such that the total of the credits of all the offered courses by the department is strictly greater than 40?

(a) Select deptid, name, sum(credits) as totalcredits from department, course where totalcredits > 40
group by deptid, name having deptid = deptno.

(b) Select deptid, name, sum(credits) as totalcredits from department, course where deptid = deptno and totalcredits > 40
group by deptid, name

(c) Select deptid, name, sum(credits) as totalcredits from department, course
group by deptid, name having deptid = deptno and totalcredits > 40;

(d) Select deptid, name, sum(credits) as totalcredits from department, course where deptid = deptno
group by deptid, name having totalcredits > 40

Ans. (d) : The correct queries to retrieve the dept id and name of the all departments that are such that the total of the credits of all the offered courses by the department is strictly greater than 40.
Select deptid, name, sum(credits) as totalcredits from department, course where deptid = deptno group by deptid, name having totalcredits > 40

95. Given query :
select s.rollno, s.name from student s where EXISTS (select c.courseid from course c Where c.deptno = 5 and EXISTS (select e.* from enrollment e where e.courseid = c.courseid and e.rollno = s.rollno)));
Which one of the following statement is correct with respect to the given query?

(a) It retrieves the roll number and names of students who have not enrolled for all course offered by department number 5.

(b) It retrieves the roll number and names of students who have enrolled for all course offered by department number 5.

(c) It retrieves the roll number and names of students who have enrolled for at least one course offered by department number 5.

(d) It retrieves the roll number and names of students who are such that there exists at least one course offered by department number 5 which is not enrolled by the student.

Ans. (c) : Given query :
select s.rollno, s.name From student
Where EXISTS (select c.courseid from course c Where c.deptno = 5 and EXISTS (select e.* from enrollment e where e.courseid = c.courseid and e.rollno = s.rollno));
The following statement is correct with respect to the given query.

- It retrievs the roll number and name of student who have enrolled for at lest one course offered by department number 5.

96. For a knapsack problem, P and W are profits and weights for a set of 7 items as given below:
P = {9, 5, 2, 7, 6, 16, 9}
W = {2, 5, 6, 11, 1, 9, 3}
Find the optimal solution for fractional knapsack problem when its maximum capacity is 15.

(a) 43 (b) 40
(c) 45 (d) 44

Ans. (b) : To solve the fractional knapsack problem, you can use the greedy algorithm First, calculate the profit to weight ratio for each item.

$$\text{Ratio } = \frac{\text{Profit(P)}}{\text{Weight(W)}}$$

Then, sort the items in descending order based on their ratio. Now, you can start adding items to the knapsack one by one until the knapsack's weight reaches its maximum capacity of 15 let's calculate the ratios.

$$\frac{P}{W} = \{4.5, 1, 0.33, 0.63, 6, 1.77, 3\}$$

P = 6+ 9 + 9 +16 = 40
W = 1 + 2 + 3 + 9 = 15 kg
So option (b) is correct.

97. For a knapsack problem, P and W are profits and weights for a set of 7 items as given below:
P = {9, 5, 2, 7, 6, 16, 9}
W = {2, 5, 6, 11, 1, 9, 3}
What is the maximum value of items you can carry if maximum weight allowed is 25 with a fractional knapsack?

(a) 48.1 (b) 79
(c) 47.9 (d) 40

Ans. (a) : Given a knapsack problem, P and W are profits and weights for a set of 7 items as
P = {9, 5, 2, 7, 6, 16, 9}
W = {2, 5, 6, 11, 1, 9, 3}
We can use the greedy algorithm first, calculate the profit - to weight ratio for each item

$$\text{Ratio} = \frac{\text{Profit(P)}}{\text{Weight(W)}}$$

$$\frac{P}{W} = \{4.5,1,0.33,0.63,6,1.77,3\}$$

Then sort the items in descending order on their ratio.
W = 1 + 2 + 3 + 9 + 5 + 5 = 25 kg

$$P = 6 + 9 + 9 + 16 + 5 + \frac{5}{11}\times 7$$

= 45 + 3.1
P = 48.1
So option (a) is correct

98. For a knapsack problem, P and W are profits and weights for a set of 7 items as given below:
P = {9, 5, 2, 7, 6, 16, 9}
W = {2, 5, 6, 11, 1, 9, 3}
If new item (P=5, W=2) is added to the list, total items is 8 and capacity is 25, then the best-case time of fractional knapsack for this case is ______

(a) O(8) (b) O(64)
(c) O(24) (d) O(56)

Ans. (c) : A knapsack problem P and W are profits and weights for a set of 7 item as given below.
P = {9, 5, 2, 7, 6, 16, 9}
W = {2, 5, 6, 11, 1, 9, 3}
new item (P=5, W=2) is added.
We can use the greedy algorithm first calculate the profit to weight ratio for each item.

$$\text{Ratio} = \frac{\text{Profit(P)}}{\text{Weight(W)}}$$

We know that
Best-case time of fractional knapsack = n logn
n is number of item
So total number of item = 8
Best-case time of fractional knapsack
= 8 log8
= 8 $\log 2^3$
= 8 ×3
= O(24)
So option (c) is correct.

99. For a knapsack problem, P and W are profits and weights for a set of 7 items as given below:
P = {9, 5, 2, 7, 6, 16, 9}
W = {2, 5, 6, 11, 1, 9, 3}
The most efficient solution for the fractional knapsack problem with maximum capacity 15 can be obtained by using ______

(a) Divide and Conquer
(b) Dynamic Programming
(c) Greedy Algorithm
(d) Backtracking

Ans. (c) : A knapsack problem, P and W are profits and weights for a set of 7 items as given below:
P = {9, 5, 2, 7, 6, 16, 9}
W = {2, 5, 6, 11, 1, 9, 3}
The most efficient solution for the fractional knapsack problem with maximum capacity 15 can be obtained by using Greedy Algorithm.
So option (c) is correct.

100. For a knapsack problem, P and W are profits and weights for a set of 7 items as given below:
P = {9, 5, 2, 7, 6, 16, 9}
W = {2, 5, 6, 11, 1, 9, 3}
Which of the following statements is correct for a knapsack?

(a) 0/1 knapsack can be applied for all kinds of problems but fractional knapsack can be applied only if the problem meets certain conditions.
(b) There is no difference between the two. They give the same solution for any problem.
(c) We use greedy algorithm for 0/1 knapsack and dynamic programming for fractional knapsack.
(d) In 0/1 knapsack problem, items cannot be divided into smaller portions and in fractional knapsack we can divide the item into required proportions of weight.

Ans. (d) :
In 0/1 knapsack problem is Dynamic Programming.
■ Fractional knapsack problem is Greedy Algorithm.
So correct statement is
In 0/1 knapsack problem, items cannot be divided into smaller portions and in fractional knapsack we can divide the item into required proportions of weight.